

INFINITY UNIVERSAL ROUTER Audit Summary

Prepared by Cyfrin

Version 3.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Pancake Swap on INFINITY UNIVERSAL ROUTER. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 3.0 dated March 10, 2025, was delivered to Pancake Swap, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 9 days, the Cyfrin team conducted an audit on the INFINITY UNIVERSAL ROUTER code provided by Pancake Swap. The findings consist of 2 Medium severity issues with the remainder being informational.

Summary

Project Name	INFINITY UNIVERSAL ROUTER
Repository	pancake-v4-universal-router
Commit	d2764d0115f1...
Audit Timeline	Sep 23 - Oct 4th, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	2	2	0
Low Risk	0	0	0
Informational	3	2	1
Gas Optimizations	0	0	0
Total	5	4	1

6 Audit Scope

Cyfrin conducted a security audit of the PancakeSwap Infinity Universal Router (previously known as the Universal Swap Router), a smart contract system designed to enable efficient token swaps across multiple protocols and versions of the PancakeSwap decentralized exchange.

Smart contracts from the `pancake-v4-core` and `pancake-v4-periphery` repositories were considered out of scope for this audit.

The following files were included in the scope of the audit:

```
src/UniversalRouter.sol
src/base/Dispatcher.sol
src/base/Lock.sol
src/base/RouterImmutables.sol
src/modules/pancakeswap/v2/V2SwapRouter.sol
src/modules/pancakeswap/v3/V3SwapRouter.sol
src/modules/pancakeswap/v4/V4SwapRouter.sol
src/modules/pancakeswap/StableSwapRouter.sol
src/modules/Payments.sol
src/modules/Permit2Payments.sol
src/modules/V3ToV4Migrator.sol
src/libraries/BytesLib.sol
src/libraries/Commands.sol
src/libraries/Locker.sol
src/libraries/MaxInputAmount.sol
src/libraries/UniversalRouterHelper.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The PancakeSwap Infinity Universal Router is a smart contract system designed to facilitate efficient token swaps across multiple versions of the PancakeSwap decentralized exchange (v2, v3, v4) and StableSwap pools. Key features include:

- Unified entry point for executing trades across different PancakeSwap protocols
- Command-based architecture allowing complex, multi-step trades in a single transaction
- Integration with Permit2 for gasless token approvals
- Support for native token wrapping/unwrapping
- Flexible routing options including multi-hop trades and split trades across multiple pools

The system utilizes a modular design with separate modules for each supported protocol (v2, v3, v4, StableSwap), allowing for easy extensibility and maintenance.

9 Attestation

Cyfrin attests that it performed a security review of the INFINITY UNIVERSAL ROUTER code provided by Pancake Swap.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-03-10-cyfrin-pancakeswap-v3.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Pancake Swap has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=3412b06d048c3743fd75254f975c2ad590d856317020ab08f7f07ac6e342f5eb
F=2025-03-10-cyfrin-pancakeswap-v3.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.